Answers and talk notes

Least Common Multiple

What your student is learning

Tonight we use the least common multiple to solve a real buying problem — the same number of two things, with none left over.

Answers

Ask your student to explain before you look.

1. **A.** the same number of each, none left over. Look for the same number of each, none left over: it means finding a number both pack sizes reach, the LCM.
2. **A.** 15. Multiples of 5: 5, 10, 15. Multiples of 3: 3, 6, 9, 12, 15. The first match is 15.
3. **B.** 3. 24 ÷ 8 = 3 packs of straws.
4. **D.** 24. Multiples of 12: 12, 24. Multiples of 8: 8, 16, 24. The first match is 24.
5. **C.** 7. 30 ÷ 6 = 5 packs of pens and 30 ÷ 15 = 2 packs of caps, so 5 + 2 = 7 packs.

Talk about it

What is the smallest number where both packs come out even? How many packs of each is that?

Home mission

Line up cups in groups of 6 on one side and spoons in groups of 8 on the other. Keep adding groups until the totals match. Which total do you reach?

When they are stuck, ask

- What numbers can you reach counting by each pack size?
- What is the first number both can reach?
- Is that the number of items, or the number of packs?

Try not to say

- Multiply the pack sizes, that is always the LCM.
- Buy the bigger number, extras do not hurt.
- The LCM is the number of packs to buy.